

# BOBICHENG ZHANG

PHD CANDIDATE · PSYCHOLOGY · COGNITIVE NEUROSCIENCE

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## EDUCATION

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### Michigan State University

PHD PSYCHOLOGY

East Lansing, MI

Dec 2022 - Present

- Dissertation: Dissecting pupillary response pattern accompanying perceptual switches during bistability

### Michigan State University

MA PSYCHOLOGY

East Lansing, MI

Jul 2020 - Dec 2022

- Thesis: Modest Effect of Knowledge on Bistable Perception during Structure-from-Motion

### University of Chicago

MA SOCIAL SCIENCES

Chicago, IL

Oct 2018 - Jun 2019

- Thesis: Ambiguity resolution with multi-frequency interocular switch rivalry

### University of Denver

BS PSYCHOLOGY

Denver, CO

Sep 2014 - Jun 2018

- Minors in Mathematics and Sociology

## RESEARCH

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### Michigan State University - Department of Psychology

ADVISOR: JAN BRASCAMP

East Lansing, MI

Aug 2020-Present

- Collect and analyze physiological data such as eye movement, pupil size, skin conductance and heart rate.
- Develop custom algorithms in Python that allow for inference of perceptual interpretations based on eye movements during bistability without explicit report.
- Present findings at various conferences such as Society for Neuroscience and Vision Sciences Society.
- Advise and support undergraduate students in literature review, statistical analysis and presenting at conference.

### Energize Colorado - Research and Insights

SUPERVISOR: DUSTIN WEILBACH

Denver, CO

Nov 2021 - Oct 2025

- Developed dozens of surveys using Qualtrics to collect information from small business owners to compute performance and health metrics.
- Performed extensive data wrangling for complex data structures to extract clearer insights and prepare for further analysis.
- Conducted exploratory analyses such as principle component analysis and parallel analysis to establish construct validity for hypothesized components of small business resilience.
- Collaborated with other teams at EC to maximize the utilization and effectiveness of actionable, data-driven insights.

### University of Chicago - Institutue for Mind and Biology

ADVISOR: STEVE SHEVELL

Chicago, IL

Oct 2018 - Jul 2020

- Reviewed literature, conducted experiments and completed a highly rated MA thesis in less than 8 months.
- Performed statistical analyses such as non-orthogonal contrast to demonstrate that binocularly driven neurons in V1 resolve neural ambiguity during chromatic interocular-switch rivalry.
- Presented findings at a graduate student conference and departmental forums.

- Independent Study: "The effect of chromatic adaptation on affective perceptual decision making"
- Reviewed literature in efforts to understand relationships between colors and emotion perception.
- Learned to program psychophysics experiments in Matlab to present morphed emotional faces that were backward-masked.

## PUBLICATIONS

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- Brascamp JW, **Zhang B**, Marshev V (2025) Fixational saccade inhibition and pupil dilation during self-paced limb movement preparation. *PLoS One* 20(10): e0335504. <https://doi.org/10.1371/journal.pone.0335504>
- Zhang, B.**, Marshev, V., & Brascamp, J. W. (2025). The pupil response to perceptual switches: What happens when you ignore them. *Journal of Vision*, 25(8), 5-5. <https://doi.org/10.1167/jov.25.8.5>
- Smith, O.M., Jeffery, B.E., Leuenberger, W., Mack, K.M, Pizzo, G, **Zhang, B** & Davis, C. Widespread lack of article accessibility policies among ecology and evolution journals. *Nat Ecol Evol* (2025). <https://doi.org/10.1038/s41559-025-02694-8>
- Zhang, B.** & Brascamp, J.W. (2022). Modest effect of knowledge on bistable perception of structure-from-motion. *Vision Research*. <https://doi.org/10.1016/j.visres.2022.108118>
- Zhang, B.**, Slezak, E., Wang, W. & Shevell, S.K.(2021). Binocularly-driven competing neural responses and the perceptual resolution of color. *Journal of Vision*, 21(10), 15-15. <https://doi.org/10.1167/jov.21.10.15>

## PRESENTATIONS

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\* *presenting author*; + *presented by lead author*

- + Brascamp, J. W., **Zhang, B.**, & Marshev, V. (July 2025). Rate of fixational saccades reveals intention to move limbs. Poster Presentation, Gordon Research Conference, South Hadley, MA
- + Cottone, M., Padula, I., Gampala, G. & Brascamp, J. W. (April 2025). Investigating the Impact of Stimulus Movement on Pupil Size Independent of Luminance. Poster Presentation, The Undergraduate Research and Arts Forum, East Lansing, MI
- + Brascamp, J. W., **Zhang, B.**, Frey, H & Marshev, V. (November 2024). Habituation of the pupil response to visual events. Poster Presentation, The Psychonomic Society Annual Meeting, New York City, NY
- \* **Zhang, B.**, Brascamp, J. W., (May 2024). Perceptual switches during structure-from-motion do not elicit pupil dilation. Poster Presentation, Vision Sciences Society, St. Pete's Beach, FL
- \* **Zhang, B.**, Brascamp, J. W., (November 2023). Inferring perceptual switches during structure-from-motion without report. Poster Presentation, The Psychonomic Society Annual Meeting, San Francisco, CA
- \* **Zhang, B.**, Brascamp, J. W., (August 2023). Separating event-related pupillary responses during perceptual bistability. Poster Presentation, Systems Vision Science Symposium, Tübingen, Germany
- \* **Zhang, B.**, Brascamp, J. W., (November 2022). Modest effect of knowledge on bistable perception of structure-from-motion. Poster Presentation, Society for Neuroscience, San Diego, CA
- \* **Zhang, B.**, Brascamp, J. W., (February 2022). Modest effect of knowledge on bistable perception of structure-from-motion. Oral Presentation, Michigan State University Graduate Academic Conference, East Lansing, MI
- \* **Zhang, B.**, Shevell, S. K., (June 2019). Binocularly-driven neural responses to chromatic interocular-switch rivalry. Oral Presentation, University of Chicago Graduate Conference, Chicago, IL

## TEACHING EXPERIENCE

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|             |                                                                                |
|-------------|--------------------------------------------------------------------------------|
| Fall 2024   | Introductory Psychology, Teaching Assistant                                    |
| Spring 2024 | Sensation & Perception   Cognitive Psychology, Teaching Assistant              |
| Spring 2023 | Sensation & Perception   Cognitive Psychology, Teaching Assistant              |
| Fall 2022   | Introductory Psychology, Teaching Assistant                                    |
| Summer 2022 | Data Analysis in Psychological Research, Instructor-of-Record                  |
| Spring 2022 | Research Design and Measurements in Psychological Research, Teaching Assistant |
| Fall 2020   | Cognitive Neuroscience, Teaching Assistant                                     |

## PROFESSIONAL EXPERIENCE

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|-----------|------------------------------------------------------------|
| 2021-2025 | Research Consultant Energize Colorado                      |
| 2018-2020 | Technical Assistant University of Chicago                  |
| 2017-2018 | Behavioral Technician Firefly Autism                       |
| 2015-2016 | Undergraduate Research Assistant Fostering Healthy Futures |

## AWARDS & HONORS

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|          |                                                                                                       |           |
|----------|-------------------------------------------------------------------------------------------------------|-----------|
| Sep 2025 | Dissertation Completion Fellowship, College of Social Sciences, Michigan State University             | \$ 9,000  |
| Jul 2025 | Spotlight: Access Torch, College of Social Sciences, Michigan State University                        |           |
| May 2025 | Graduate Educator Advancement and Teaching Fellowship, The Graduate School, Michigan State University | \$ 4,500  |
| Mar 2025 | Educator of the Month, The Graduate School, Michigan State University                                 |           |
| Sep 2023 | Conference Award, Council of Graduate Students, Michigan State University                             | \$ 300    |
| Mar 2023 | Research Enhancement Award, College of Social Sciences, Michigan State University                     | \$ 660    |
| Jun 2018 | MAPSS Scholarship, Division of Social Sciences, The University of Chicago                             | \$ 24,000 |
| Nov 2017 | Public Pathways Scholar, Dept. of Psychology, University of Denver                                    |           |
| May 2017 | Academic Achievement and Citizenship Award, Dept. of Psychology, University of Denver                 |           |
| May 2017 | Academic Achievement Award, Dept. of Psychology, University of Denver                                 |           |

## OUTREACH & PROFESSIONAL DEVELOPMENT

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### SERVICE AND OUTREACH

|              |                                                                               |
|--------------|-------------------------------------------------------------------------------|
| 2024-Present | Graduate Student Accessibility and Support Network, Vice President            |
| 2025         | Fresh Check Day, Planning Committee Member                                    |
| 2025         | Neurodiversity Fair, Planning Committee Member & Student Speaker              |
| 2025         | University Health and Well-being Summit, Student Panelist                     |
| 2024-2025    | Disability Summit, Planning Committee Member & Student Speaker                |
| 2023-2025    | Graduate Academic Conference, Judge                                           |
| 2022-2024    | Psychology Scholars Undergraduate Program Workshop, Graduate Student Panelist |
| 2022-2024    | Council of Graduate Students, Student Representative                          |
| 2022-2024    | University Committee on Graduate Studies, Graduate Student Representative     |

### DEVELOPMENT

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|----------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------|
| May - Dec 2025 | GREAT Fellowship: Graduate Resources for Accommodations and Supportive Practices, Michigan State University Graduate School | East Lansing, MI  |
| Aug 2023       | Systems Vision Science Summer School & Symposium, Max Planck Institute for Biological Cybernetics                           | Tübingen, Germany |
| Jun 2022       | Certificate for College Teaching Workshop, Michigan State University                                                        | East Lansing, MI  |